

Topic: Polynomial Representation and Addition - Self-Learning

Q. Explain how a Singly Linked List can be used to represent a polynomial. Describe the algorithm to add two polynomials represented using linked lists with a suitable example.

****Definition to Remember****

A polynomial is a non-zero memory

Polynomial Representation & Addition (Self-Learning)

Each node in the linked list represents one term and is divided into three parts:

1. Coef

Structure in C:

struct P

Example Diagram for $5x^3 + 4x^1 + 2x^0$:

text

2. Polynomial Addition Algorithm

To add `Poly1` and `Poly2`, traverse both lists simultaneously using pointers `p1` and `p2`. Compare their exponents:

* **Case 1: `p1->exponent > p2->exponent`**

Append

After the loop, append any remaining terms from either `p1` or `p2` to the Result list.

3. Example Trace

* **Poly1:** $3x^2 + 5x + 2$

* **Po

4. Advantages

* **Memory Efficiency:** Only non-zero terms are stored. (An array for $x^{100} + 1$ requires 101 slots; a linked list requires only 2 nodes).

* **Dy**

Must-Write Points (for 10 marks)

1. Array

Quick Recall

Poly LL
== exp2